Statistical Power

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Introduction

A committee, a reviewer, an IRB, or a funder asks: **how many subjects do you need?**

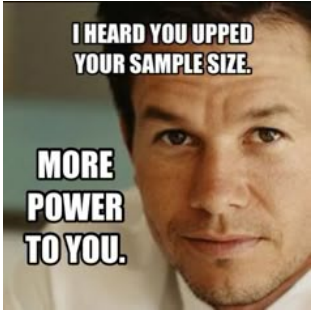


Figure 1: All problems are solved with more power.

Schedule

Min	Block	Who is working
10	What power is	Lecture
9	The formula, worked by hand; the smallest detectable effect	Lecture
16	**Your turn 1:** a tutoring study, by hand and in the EGAP app	**You work**
6	Where the assumptions come from	**Pairs**
4	The formula in R, swept over N	Demo
15	**Your turn 2:** a sleep study, on a budget	**Groups**
5	*Break*	—
14	Simulation: the recipe, checked against the formula	Demo + predictions
5	Adding a covariate	Demo + predictions
14	**Your turn 3:** simulate a study with dropout	**Groups**
3	What invalidates a standard calculation	Lecture
18	**Your turn 4:** simulate your own design	**You code**

Learning Objectives

By the end of the session you should be able to:

You will be able to	You will show it by
Define power in terms of Type I and Type II error, and **name** the quantities that determine it	Saying which of those quantities a design lets you change
Compute power and the minimum detectable effect for a two-arm difference in means	Working an example by hand, then checking it in the EGAP calculator
Sweep the formula over sample sizes and effect sizes in R, and **justify** the effects you assume	Finding the N a study needs, and whether a budget can pay for it
Simulate power for a design the formula does not cover	Writing `generate()`, `analyze()`, and `success()` for your own study
Identify design features that invalidate a standard calculation	Naming the adjustment for clustering, attrition, take-up, and interactions

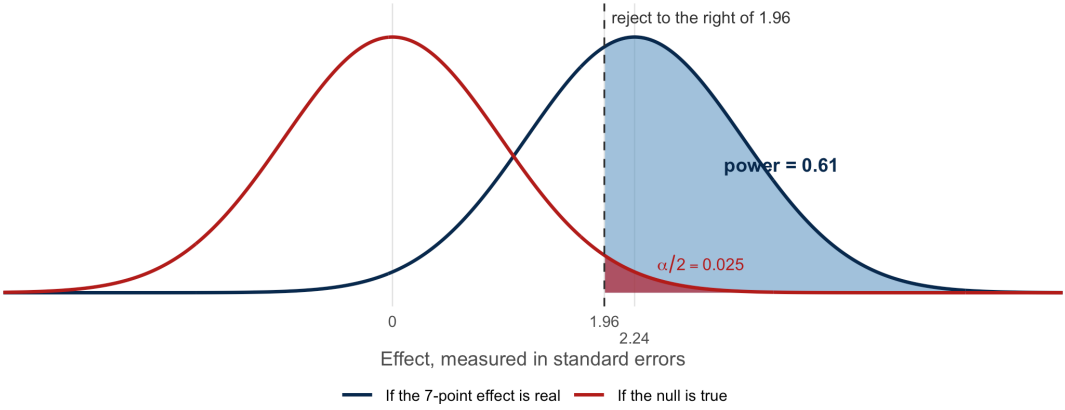
False Negatives and False Positives (Type I and Type II Errors)

- A hypothesis test can make two errors:
  - *False positive* (α, Type I): claiming an effect is *present* when the **null hypothesis is true**.
  - *False negative* (β, Type II): claiming an effect is *absent* when the **null hypothesis is false**.
- Power is the probability of avoiding a false negative: power $= 1 - \beta$.
- The usual target is power ≥ 0.80, a 20% chance of missing a real effect. Like $\alpha = 0.05$, this is a convention; use a different value if you can justify it.



Figure 2: Dr. Jacob Cohen (1923-1998)

What Power Looks Like



Red is the distribution of the test statistic if there is no effect; the shaded area past 1.96 is α, the false-positive rate. **Blue** is the distribution if the effect is real; **the shaded area past the same line is power**.

Type S and Type M Errors

- Underpowered studies do more than miss real effects: when they *do* find a significant effect, that estimate is likely to be wrong.
- A significant estimate can be wrong in two ways:
  - *Type S (sign)*: the estimate has the opposite sign of the true effect. At 6% power, 24% of significant estimates do.
  - *Type M (magnitude)*: the estimate overstates the true effect. At 6% power, significant estimates are 9.5 times too large on average.

Source: Gelman and Carlin (2014), *Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors.*

Core Notation

Power depends on four quantities:

Symbol	Quantity	What it is	Can you change it?
N	Sample size	Total number of subjects.	Yes, within your budget.
$\mu_t - \mu_c$	Effect size	The true difference between treatment and control means.	Partly: dose, duration, framing.
σ	Standard deviation	How much outcomes vary between subjects.	Partly: better measures, covariates.
α	Significance level	The false-positive rate you accept.	Yes; usually 0.05.

Larger effects, smaller σ, and larger N all raise power. In code and in the appendix we write the effect as δ.

The Formula for a Two-Arm Experiment

$$\text{power} \;=\; \Phi\left(\frac{|\mu_t - \mu_c|\sqrt{N}}{2\sigma} \;-\; \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)\right)$$

Φ is the standard normal CDF: $\Phi(x)$ is the probability that a standard normal variable is at or below x. Φ^{-1} is its inverse: it returns the z value with a given area to its left.

This particular formula holds when:

- subjects are randomly assigned to two arms of $N/2$ each,
- the outcome is normally distributed with the same σ in both arms,
- the analysis is a two-sided difference-in-means test.

EGAP writes power as β; we use β for the Type II error rate, so power $= 1 - \beta$.

Source: Coppock, *10 Things to Know About Statistical Power* (EGAP), originally found in Gerber & Green (2012), *Field Experiments: Deisgn, Analysis, and Interpretation.*

A Running Example: Does the Message Move Opinion?

You are fielding a survey experiment with **164 respondents**, randomly assigned 50/50 to read a short informational message (treatment) or no message (control). Support for the policy is measured on a 0–100 feeling thermometer, and a pilot study supplies your assumptions.

Symbol	Value	Where it comes from
N	164	Total respondents (82 per arm).
μ_t	57	Assumed mean support under treatment.
μ_c	50	Control mean, from the pilot.
σ	20	Standard deviation, from the pilot.
α	0.05	Two-sided, by convention.

Assumed effect: $\mu_t - \mu_c = 7$ points, or $0.35 \times \sigma$. **What is the power to detect a 7-point effect with 164 respondents?**

Step 1: Do the Arithmetic

$$\text{power} = \Phi\left(\frac{|\mu_t - \mu_c|\sqrt{N}}{2\sigma} - \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)\right) \qquad \text{the formula}$$

$$= \Phi\left(\frac{7\sqrt{164}}{2(20)} - 1.96\right) \qquad \text{plug in;} \quad \Phi^{-1}(0.975) = 1.96$$

$$= \Phi(\mathbf{2.24} - 1.96) = \Phi(\mathbf{0.28}) \qquad 7 \times 12.81/40 = 2.24$$

$\underbrace{\mathbf{2.24}}_{\text{expected } z}$ $\underbrace{\mathbf{1.96}}_{\text{critical value}}$ $\underbrace{\mathbf{0.28}}_{\text{difference}}$

The smaller the difference between the expected z and the critical value, the lower the power.
(The appendix shows how to find 1.96 in the z-table.)

Step 2: Read the Answer Off the Table

$$\text{power} = \Phi(\mathbf{0.28})$$

Split the z-score across the two margins of the table:

- **0.2** selects the **row**
- **0.08** selects the **column**

The cell where they meet is the area to the *left* of z, which is the power.

z	.00	.01	.02	.03	.04	.05	.06	.07	**.08**	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	**.6103**	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224

$$\text{power} = \Phi(0.28) = \mathbf{0.61}$$

The Smallest Effect This Study Can Detect

Fix power at 80% and solve the same formula for the effect. The result is the **minimum detectable effect** (MDE).

$$\text{MDE} = \underbrace{\left(\Phi^{-1}\left(1-\tfrac{\alpha}{2}\right)+\Phi^{-1}(0.80)\right)}_{1.96+0.84=2.80} \times \underbrace{\frac{2\sigma}{\sqrt{N}}}_{\text{standard error}}$$

N	Minimum detectable effect	In standard deviations
164	**8.75 points**	**0.44σ**
257	6.99	0.35σ
400	5.60	0.28σ
800	3.96	0.20σ
1396	3.00	0.15σ

The pilot assumes 7 points, but with 164 respondents only effects of **8.75** points or more have 80% power.

Solved for N instead: $N = 4\sigma^2\left(z_{1-\alpha/2}+z_{0.80}\right)^2/(\mu_t-\mu_c)^2$.

Your Turn: Is the Tutoring Program Worth Studying?

A colleague is evaluating a new tutoring program. **400 students** will be randomly assigned, 200 to tutoring and 200 to business as usual, with final exam score as the outcome. Students in previous years average a final exam score of **72** points with a standard deviation of **10**, and the program is expected to add **3** points.

What is the power of this design?

Symbol	Value
N	400
μ_t	75
μ_c	72
σ	10
α	0.05, two-sided

The formula and the z-table are on your **handout**.

Work it in three steps:

1. Compute the expected z
2. Subtract the critical value
3. Look the difference up in the table

If you finish early: what is the MDE at $N = 400$?

Your Turn: The Solution

$$\text{power} = \Phi\left(\frac{|75-72|\sqrt{400}}{2(10)} - \Phi^{-1}\left(1-\tfrac{0.05}{2}\right)\right) \quad \textbf{Step 1:} \text{ plug in}$$

$$= \Phi\left(\frac{3\times 20}{20} - 1.96\right) \quad \textbf{Step 2:} \ \sqrt{400}=20;\ \Phi^{-1}(0.975)=1.96$$

$$= \Phi(3.00-1.96) = \Phi(\mathbf{1.04}) \quad \textbf{Step 3:} \text{ simplify}$$

$$= \mathbf{0.85} \quad \textbf{Step 4:} \text{ row 1.0, column .04} \rightarrow .8508$$

MDE at $N = 400$: $\quad 2.80 \times 2(10)/\sqrt{400} = \mathbf{2.80}$ points.

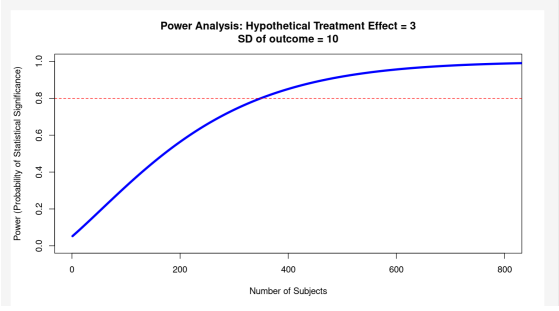
Check Your Work: The EGAP Power Calculator

https://egap.shinyapps.io/power-app/

Our symbol	Field in the app	Enter
$\mu_t - \mu_c$	Treatment Effect Size	3
σ	Standard Deviation of Outcome Variable	10
α	Significance Level	0.05
	Power Target	0.8
	Maximum Number of Subjects	800

Leave *Clustered Design* and *Binary Dependent Variable* unchecked. (The appendix shows what clustering does to N.)

Reading the Power Curve



- The x-axis is **total sample size**; the y-axis is **power**.
- The blue curve shows power as N grows, holding the effect at 3 and σ at 10.
- The red dashed line marks the **80% target**. The curve crosses it at the N you need: **349**.
- Our $N = 400$ is to the *right* of that point. Reading up from 400 gives $\approx$ **0.85**.
- The curve flattens: past about 600 students, each additional student adds little power.

Where Do the Assumptions Come From?

Power calculations are most useful *before* data collection, so every input is an estimate. **In pairs:** pick a study one of you is planning. For each row, say where your number would come from.

The estimate	**Possible sources**
How big an effect should I expect? $\mu_t - \mu_c, \quad \sigma$	
How many people can I afford to treat? N	
How many will respond and stay? N analyzed	

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How many will respond and stay? N analyzed	

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Each source gives a range, not a single number. The next step is to compute power across that range.

What If We Don't Know the Effect?

Back to the survey experiment. The pilot fixed everything *except* the expected effect size.

Fixed by the pilot	
μ_c	50
σ	20
α	0.05
target power	0.80
$\mu_t - \mu_c$	?
N	?

Source of the effect	$\mu_t - \mu_c$
Our own pilot	7
A published study of a similar message	5
That published effect, discounted for publication bias	3

Instead of choosing one, compute power for all three across a range of N.

The Formula, in Code

$$\text{power} = \Phi\left(\frac{|\mu_t - \mu_c|\sqrt{N}}{2\sigma} - \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)\right)$$

```r
power_calc <- function(mu_t, mu_c, sigma, N, alpha = 0.05) {
  z <- abs(mu_t - mu_c) * sqrt(N) / (2 * sigma)
  cv <- qnorm(1 - alpha / 2)
  pnorm(z - cv) + pnorm(-z - cv) # both rejection tails
}
```

Φ = `pnorm` $\qquad$ Φ^{-1} = `qnorm`

Power Across a Range of N

```r
grid <- expand.grid(N = seq(60, 1600, by = 10), delta = c(3, 5, 7))

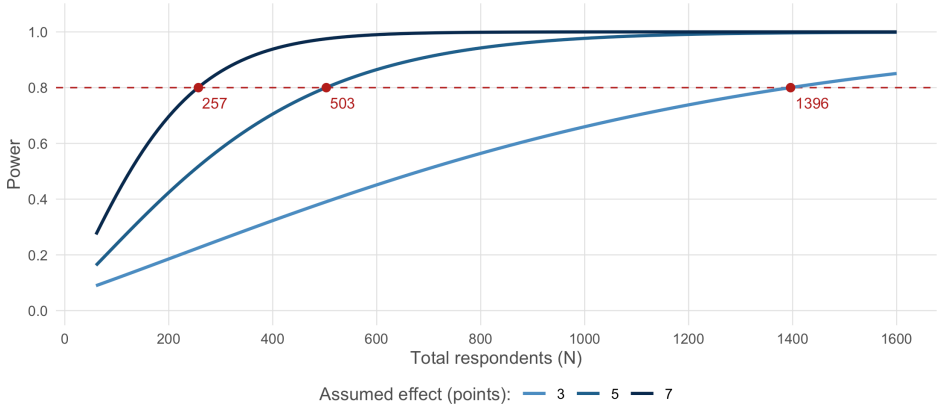
grid$power <- power_calc(
  mu_t = 50 + grid$delta,
  mu_c = 50,
  sigma = 20,
  N = grid$N
)

head(grid, 3)
```

```
   N delta      power
1 60     3 0.08947303
2 70     3 0.09618740
3 80     3 0.10293349
```

Three effect sizes $\times$ 155 values of N = 465 rows. Next, plot `power` against `N`.

How Many Respondents Do We Need?



Your Turn (Groups): A Sleep Workshop on a Budget

The campus wellness office wants to test a two-session **sleep workshop** for undergraduates. Students are randomized to the workshop or a waitlist and wear a sleep tracker for two weeks afterward. The outcome is average nightly sleep, in minutes.

From last year's tracker data	
μ_c	400 minutes
σ	60 minutes
α	0.05
Budget	
Cost per student	\$40
Total	\$12,000

Three sources disagree about the effect:

Source	**Effect**
A pilot with 20 students	+25 min
A meta-analysis of similar brief workshops	+15 min
The office director: "Under 10 minutes isn't worth running the workshop"	+10 min

All numbers are hypothetical.

Your Turn (Groups): What to Produce

Work in groups of 3–4. You have about 12 minutes. Open `scripts/sleep_group_activity.R`.

1. **Discuss:** which source would you trust most, and what is wrong with each one?
2. **Sweep:** choose the effect sizes to plan for, and find the smallest N that reaches 80% power for each.
3. **Budget:** how many students does $12,000 pay for? Which of your effects can that N detect?
4. **Report:** write the sentence your group would put in a proposal: *"With N = ___, the study has 80% power to detect effects of ___ minutes or larger."*

```
grid <- expand.grid(N = seq(___, ___, by = 10),
                    delta = c(___, ___, ___))
grid$power <- power_calc(mu_t = mu_c + grid$delta, mu_c = mu_c,
                         sigma = sigma, N = grid$N)
```

Your Turn (Groups): One Solution

```
grid <- expand.grid(N = seq(50, 1500, by = 10), delta = c(10, 15, 25))
grid$power <- power_calc(400 + grid$delta, 400, sigma = 60, N = grid$N)

# smallest N with 80% power, for effects of 10, 15, and 25 minutes
sapply(c(10, 15, 25), \(d) min(grid$N[grid$delta == d & grid$power >= 0.8]))
```

```
[1] 1140  510  190
```

```
mde_300 <- (qnorm(0.975) + qnorm(0.80)) * 2 * 60 / sqrt(300) # $12,000 / $40
round(mde_300, 1)
```

```
[1] 19.4
```

The budget pays for 300 students, enough to detect effects of 19.4 minutes or more. That covers the pilot's 25 minutes, but a pilot of 20 students gives a noisy estimate. The meta-analysis's 15 minutes needs about 510 students ($20,400).

Break

5 minutes

Where the Formula Stops Applying

The formula covers one design: two arms, a normal outcome, a difference in means. Other tests have their own formulas, and many designs have none.

The formula assumes	So it does not cover
Two arms, split 50/50	Three arms, unequal allocation
One continuous, normal outcome	Binary outcomes, counts, skewed scales
A simple difference in means	Covariates, fixed effects, IV, mixed models
Independent observations	Clustered or repeated measures
Everyone recruited is analyzed	Attrition, non-response
A significance test	Bayesian or estimation-based goals

A More General Definition of Power

Power is the probability that the analysis **recovers** a real effect, if one exists.

- So far, "recovers" has meant $p < 0.05$.
- In a simulation, it can mean any criterion you can check in code: the estimate is within a set distance of the truth, the confidence interval excludes zero, a posterior probability exceeds 0.95.
- With this definition, power can be estimated for any design whose data you can simulate.

What should we do in those other cases?

What would it mean then, to approach asking questions about power or sample size in a more general sense where we may want to consider:

- Non-parametric approaches
- Bayesian approaches
- Or even just moderately more complicated study designs?

Well, we go gambling



Figure 3: Oxford Casino Hotel

An aside into the casinos at Monaco

- Monte Carlo refers to a broad class of methods wherein you use repeated random sampling to obtain some form of numerical result
- It will be very useful for us when we cannot rely on a closed form solution (which, as a reminder, is in most cases!)
- Essentially, we can use randomness to solve deterministic problems

For any problem with probabilistic interpretation, it can be solved with Monte Carlo



Figure 4: Primary developer of the MC method, Stanisław Ulam, who was inspired by his uncle's gambling

Monte Carlo Simulation: A Roulette Example

Let's imagine we want to know the answer to a gambling-related problem with a closed-form answer:

What is the probability of winning on red three spins in a row?

- A roulette wheel has 38 pockets, 18 of them red.
- Each spin is independent, so

$$P(\text{3 reds in a row}) = (18/38)^3 \approx 0.106$$

- Next: estimate the same number by simulating spins, and compare.



Figure 5: A standard roulette table

Monte Carlo Simulation: Simulating the Spins

But we could also just... go to the casino 100,000 times:

Let's simulate 100,000 sets of three spins and count how often all three are red:

```
set.seed(2026)
spins <- matrix(runif(3 * 100000) < 18 / 38, ncol = 3) # each row is 3 spins
mean(rowSums(spins) == 3) # proportion of rows where all three are red
```

```
[1] 0.1059
```

If you can write down how the data are generated, you can estimate a probability by running that process many times and counting. Power is a probability, so it can be estimated the same way.

So, how do we do a Monte Carlo simulation, anyways?

In the parametric context, a Monte Carlo power analysis looks like this:

1. **Generate** random data from some distribution with the properties that you expect your real data to have
2. **Analyze** this randomly generated data set with your statistical analysis, whatever that is
3. **Store** the value(s) of the analysis that you're interested in
4. **Repeat** the above steps many times
5. **Compute** power by looking at the proportion of times the statistic or value you're interested in was "correct"

This can be arbitrarily complex or simple!

The Simulation Recipe

Write three functions:

- `generate()`: make one fake data set from the model you expect your data to follow
- `analyze()`: run the analysis you will run on the real data
- `success()`: return `TRUE` if this run recovered the effect

Then run them many times. The proportion of successes is the estimated power:

```
simulate_power <- function(n_sims, generate, analyze, success) {
  hits <- replicate(n_sims, success(analyze(generate())))
  mean(hits) # proportion of successful runs = power
}
```

Generating Data for the Tutoring Study

We need to make a statement on how we think the data are being generated. For student i:

$$\text{post}_i = \underbrace{\mu_c}_{\text{control mean}} + \underbrace{\delta T_i}_{\text{tutoring effect}} + \underbrace{\varepsilon_i}_{\text{noise}}, \qquad \varepsilon_i \sim N(0, \sigma^2)$$

$\mu_c = 72$; ; $T_i = 1$ if student i was tutored, 0 if not; ; $\delta = 3$; ; $\sigma = 10$. We use $N = 350$, close to the 349 the EGAP curve gave for 80% power.

```
gen_basic <- function(n = 350, effect = 3, sd = 10) {
  treat <- rep(0:1, each = n / 2) # half control, half tutored
  post <- 72 + effect * treat + rnorm(n, mean = 0, sd = sd)
  data.frame(treat, post)
}
```

Analyzing It and Defining Success

```r
analyze_lm <- function(dat) {
  fit <- lm(post ~ treat, data = dat) # same as an equal-variance t-test
  summary(fit)$coefficients["treat", ] # estimate, SE, t, p for treat
}

success_sig <- function(est) {
  est[["Pr(>|t|)"]] < 0.05 # did we reject the null?
}

analyze_lm(gen_basic())
```

```
  Estimate Std. Error    t value   Pr(>|t|)
3.0384590  1.1588225  2.6220228  0.0091249
```

Predict: the formula gives power $= 0.8$ at $N = 350$. What will 2,000 simulated studies give?

Does the Simulation Agree with the Formula?

```
set.seed(2026)
simulate_power(n_sims = 2000, generate = gen_basic,
               analyze = analyze_lm, success = success_sig)
```

```
[1] 0.8105
```

- Pretty close for only 2000 simulations!

The formula gives 0.8. The simulated estimate has its own error. With S simulations, the Monte Carlo standard error is

$$\text{SE} = \sqrt{\frac{p(1-p)}{S}}$$

which is about 0.009 for $p = 0.80$ and $S = 2{,}000$. For a more precise estimate, increase S.

Adding some complication – a pre-test

Schools often have each student's **pre-test score**. The model for student i becomes:

$$\text{pre}_i \sim N(70, \sigma^2)$$

$$\text{post}_i = \mu_c + \underbrace{\rho\,(\text{pre}_i - 70)}_{\text{prior achievement}} + \delta\, T_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0,\ \sigma^2(1-\rho^2))$$

- $\rho = 0.6$ is the correlation between pre-test and end-of-term scores.
- The total standard deviation of post-test scores is still $\sigma = 10$.

Predict: at the same $N = 350$, does adjusting for the pre-test raise or lower power? By how much?

Adjusting for the Pre-Test: Code

Only `generate()` and `analyze()` change:

```r
gen_pretest <- function(n = 350, effect = 3, sd = 10, rho = 0.6) {
  treat <- rep(0:1, each = n / 2)
  pre <- rnorm(n, mean = 70, sd = sd)
  post <- 72 +
    rho * (pre - 70) +
    effect * treat +
    rnorm(n, mean = 0, sd = sd * sqrt(1 - rho^2))
  data.frame(treat, pre, post)
}

analyze_ancova <- function(dat) {
  summary(lm(post ~ treat + pre, data = dat))$coefficients["treat", ]
}
```

Adjusting for the Pre-Test: Result

```
set.seed(2026)
simulate_power(2000, gen_pretest, analyze_lm, success_sig)     # no pre-test
```

```
[1] 0.807
```

```
simulate_power(2000, gen_pretest, analyze_ancova, success_sig) # pre-test
```

```
[1] 0.9415
```

- Same students, same N: power rises from about 0.80 to about 0.94.
- Power depends on the **analysis** as well as on N. Adjusting for a pre-treatment covariate costs no additional students.
- If we were really strapped for students to enroll, this could be really helpful to know!

(To be fair: this case still has a closed form (σ just becomes $\sigma\sqrt{1-\rho^2}$) but we didn't need to know that!)

Your Turn (Groups): The Tutoring Study with Dropout

Past studies show that students who are doing *badly* stop attending, and the **control group loses more** students (about 23% vs. 11%). Dropout depends on the outcome, so it changes the sample size, the estimate, and the noise together. There is no standard formula for this design. Open `scripts/dropout_group_activity.R`; it contains this function and the ones from the demo.

```
gen_dropout <- function(n = 350, effect = 3, sd = 10, rho = 0.6) {
  dat <- gen_pretest(n, effect, sd, rho)
  p_drop <- plogis(-2.2 + 0.7 * (dat$treat == 0) - 0.12 * (dat$post - 72))
  dat[runif(n) > p_drop, ] # we only observe the students who stayed
}
nrow(gen_dropout()) # out of the 350 we enrolled
```

```
[1] 289
```

Your Turn (Groups): Tasks

Work in groups of 3–4. You have about 10 minutes.

1. **Predict** before running anything: will dropout raise or lower power? Will the average estimate be above or below the true effect of 3? Why?
2. **Estimate power** with and without dropout, using `simulate_power()`.
3. **Check for bias:** average the estimated effect over 1,000 simulated studies with dropout.
4. **Redefine success:** write a `success()` that counts a run as a success if the estimate is within 1.5 points of 3. Compare power with and without dropout.
5. *If time:* **sweep over** N from 150 to 500. Where does power reach 80%? Does enrolling more students remove the bias?

Use 500–2,000 simulations per estimate. Each task in the script has blanks (`___`) to fill in.

Your Turn (Groups): Solution, Power and Bias

```
set.seed(2026)
simulate_power(2000, gen_dropout, analyze_ancova, success_sig)
```

```
[1] 0.687
```

```
estimates <- replicate(2000, analyze_ancova(gen_dropout())[["Estimate"]])
mean(estimates) # the true effect is 3
```

```
[1] 2.275113
```

- Low-scoring control students leave, which raises the observed control mean, so the estimate is biased **downward**.
- Power falls because of the smaller sample *and* the smaller estimated effect.
- Enrolling more students raises power, but does not remove the bias (see the sweep two slides ahead).

Your Turn (Groups): Solution, Other Definitions of Success

We know the true effect is 3, so success can be defined by the estimate instead of the p-value:

```
success_close <- function(est) abs(est[["Estimate"]] - 3) < 1.5

set.seed(2026)
simulate_power(2000, gen_pretest, analyze_ancova, success_close) # no dropout
```

```
[1] 0.9185
```

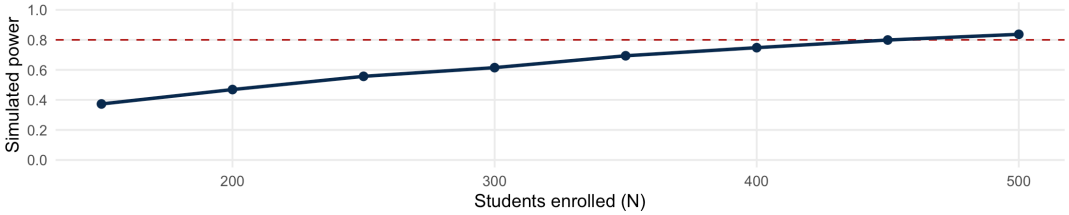
```
simulate_power(2000, gen_dropout, analyze_ancova, success_close) # dropout
```

```
[1] 0.786
```

Other options: the CI excludes zero; the CI is narrower than a set width; the estimate has the correct sign; a posterior probability exceeds 0.95. At low power, significant estimates often have the wrong sign or overstate the effect several-fold (Type S and M errors; Gelman and Carlin, 2014).

Your Turn (Groups): Solution, How Many Students?

```
set.seed(2026)
ns <- seq(150, 500, by = 50)
powers <- sapply(ns, \(n) {
  simulate_power(1000, generate = \() gen_dropout(n = n),
                 analyze = analyze_ancova, success = success_sig)
})
```



Power reaches 80% at about 450 students, but more students do not make the estimate more accurate.

What Invalidates a Standard Calculation

The formula assumes you randomize individuals, measure a continuous outcome, and analyze everyone you recruit. If any of these fails, the required N changes, usually by a lot.

If your design has...	**Adjust N by**	**Survey example ($N = 257$ for 7 points)**
Groups randomized, not individuals	$\times$ DEFF $= 1 + (m-1)\rho$	Groups of 25, $\rho = 0.15$: 257 $\rightarrow$ **1,183**. Analyzed as individuals, the false-positive rate is over **35%**.
Take-up below 100%	$\div$ take-up^2	50% take-up: 257 $\rightarrow$ **1,028**
Attrition	$\div$ (1 $-$ loss)	20% loss: recruit **322** to analyze 257
A binary outcome	use $\sigma = \sqrt{p(1-p)}$	30% $\rightarrow$ 42% support with $N = 164$: power **0.36**, vs. 0.61 for the continuous outcome
A subgroup difference or interaction	$\approx \times 4$	A 6-point interaction: 257 $\rightarrow$ **about 1,400**

Attrition that depends on the outcome, as in the dropout example, also biases the estimate. Simulate it.

Your Turn: Simulate Your Own Design

Open `scripts/power_sim_template.R` and fill in each `___`, running each step before moving on.

Choose your study:

- **Your own study.** Use your estimates for the control mean, effect, and standard deviation. If you have pilot or baseline data, use its mean and standard deviation.
- **No study in mind?** Use the blood-pressure coaching study on the next slide.

Complete steps 1–6 in class (about 15 minutes). Step 7 is optional if you finish early.

Mixed models are slow inside `replicate()`. Start with 200 simulations, and increase the number once your code works.

Your Turn: A Study to Simulate

A primary-care clinic wants to test a **home blood-pressure coaching program**. Patients with hypertension are randomized to six months of phone coaching or usual care. The outcome is systolic blood pressure (mmHg) at six months.

From the clinic's records	
Mean under usual care	140
Standard deviation	15
Expected effect	−5
Patients the clinic can enroll	250

All numbers are hypothetical.

Steps 1–6: Does the clinic have 80% power with 250 patients? If not, how many does it need?

Step 7, pick one complication:

- **Baseline blood pressure** is recorded at enrollment and correlates 0.5 with the six-month reading. Does adjusting for it get the clinic to 80%?
- **Unequal groups:** the clinic has coaches for only one in three patients. How much power does a 1:2 split cost at $N = 250$?

Your Turn: Questions to Answer at Each Step

1. **Inputs:** what are your control mean, effect, and standard deviation, and where did each number come from?
2. `generate()`: does the fake data set look like data your study could produce?
3. `analyze()`: is this the analysis you will run on the real data?
4. `success()`: what counts as recovering the effect?
5. **Estimate power.** Set the effect to zero: does a significance-based `success()` succeed about 5% of the time?
6. **Sweep over N.** Where does your curve cross 80%?
7. *(Optional)* **Add a complication:** unequal groups, a covariate, a skewed outcome, or dropout. How much power do you gain or lose?

Appendix: Where 1.96 Comes From

$$\Phi^{-1}(\mathbf{0.975})$$

Φ^{-1} uses the table in reverse. Given an area, find the z that produces it:

1. Find .9750 in the **body**
2. Read off its **row and column**

Row 1.9 plus column .06 gives **1.96**.

z	.00	.01	.02	.03	.04	.05	**.06**	.07	.08	.09
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	**.9750**	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857

This is Step 2 in reverse: start in the body of the table and read outward.

$\Phi^{-1}(0.975) = \mathbf{1.96}$: the critical value for a two-sided test at $\alpha = 0.05$.

Appendix: How Many Simulations Is Enough?

A simulated power estimate has standard error $\sqrt{\text{power}(1-\text{power})/S}$.

S	$\pm$ **at power** $= 0.80$	**Suitable for**
200	0.028	Checking that the code runs
1,000	0.013	Comparing designs
2,000	0.009	Teaching and prototyping
10,000	0.004	A number reported in a grant or pre-registration

A quick check: rerun with a different seed. If the estimate changes more than you can accept, increase S.

Appendix: Estimand and Estimator

	What it is	Examples
Estimand	The quantity you want to learn. A property of the *population*.	The ATE; the effect among women; an interaction; a LATE
Estimator	The procedure you run on data. A property of your *code*.	Difference in means; OLS with covariates; 2SLS; a mixed model

Power depends on the estimand, the estimator, and the design together. In the pre-test example, changing the estimator (adding a covariate) raised power. On the next slide, changing the estimand (an interaction instead of the average effect) lowers it.

Appendix: Power for an Interaction

The message moves group $g = 1$ by 10 points and group $g = 0$ by 4: the average effect is 7 and the **interaction** is 6. The interaction's standard error is twice the ATE's, so 80% power needs about **1,400** respondents, against 257 for the ATE.

```
set.seed(11)
p <- replicate(2000, {
  z <- sample(rep(0:1, each = 82))
  g <- rep(0:1, 82)
  y <- rnorm(164, 50 + 4 * z + 6 * z * g + 2 * g, 20)
  c(ate = coef(summary(lm(y ~ z)))["z", 4],
    int = coef(summary(lm(y ~ z * g)))["z:g", 4])
})
rowMeans(p < 0.05) # N = 164
```

```
  ate   int
0.605 0.149
```

Appendix: What If x Is a Confounder?

Here **z** is **not randomized**: higher **x** makes treatment more likely. The unadjusted model rejects almost every time, but its estimate is far from 7. A rejection rate measures power only when the estimator is unbiased.

```
set.seed(3)
sim_obs <- function(adjust, N = 164) {
  out <- replicate(2000, {
    x <- rnorm(N)
    z <- rbinom(N, 1, plogis(1.2 * x)) # z DEPENDS on x
    y <- 50 + 7 * z + 12 * x + rnorm(N, 0, 16) # truth is still 7
    fit <- if (adjust) lm(y ~ z + x) else lm(y ~ z)
    c(coef(fit)["z"], coef(summary(fit))["z", 4])
  })
  c(estimate = mean(out[1, ]), rejection = mean(out[2, ] < 0.05))
}
rbind(`y ~ z` = sim_obs(FALSE), `y ~ z + x` = sim_obs(TRUE))
```

```
           estimate rejection
y ~ z     18.207711      1.00
y ~ z + x  7.139331      0.71
```

Appendix: When You Randomize Groups, Not People

If you assign whole classrooms, clinics, or villages, observations within a group are correlated. The required N grows by the **design effect**:

$$\text{DEFF} = 1 + (m - 1)\,\rho \qquad m = \text{cluster size}, \quad \rho = \text{intra-cluster correlation}$$

Survey experiment, 7-point effect	**Respondents needed**
Randomize individuals	257
Randomize groups of 25, $\rho = 0.15$ (DEFF 4.6)	**1,183**

This is what the *Clustered Design* option in the EGAP app accounts for.

Appendix: Clustering in Simulation

300 respondents in 12 groups of 25, $\rho = 0.15$. Analyzed as individuals, the test rejects a true null far more than 5% of the time. Analyzed as 12 group means, it holds 5%, but has little power.

```
set.seed(7)
sim_clus <- function(tau, k = 12, m = 25) rowMeans(replicate(1500, {
  zc <- sample(rep(0:1, each = k / 2)) # assign GROUPS, not people
  u <- rnorm(k, 0, 20 * sqrt(0.15)) # group-level shock
  y <- rnorm(k * m, rep(50 + tau * zc + u, each = m), 20 * sqrt(0.85))
  c(as_individuals = t.test(y ~ rep(zc, each = m))$p.value,
    as_groups = t.test(tapply(y, rep(1:k, each = m), mean) ~ zc)$p.value)
}) < 0.05)
rbind(`no effect` = sim_clus(0), `7-point effect` = sim_clus(7))
```

```
               as_individuals as_groups
no effect           0.4000000     0.050
7-point effect      0.7166667     0.238
```

References

Coppock, A. *10 Things to Know About Statistical Power*. Evidence in Governance and Politics (EGAP). https://egap.org/resource/10-things-to-know-about-statistical-power/

Gelman, A., and Carlin, J. (2014). Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors. *Perspectives on Psychological Science*, 9(6), 641–651. https://doi.org/10.1177/1745691614551642

Gerber, A. S., and Green, D. P. (2012). *Field Experiments: Design, Analysis, and Interpretation*. New York: W. W. Norton.